CONSTRUCTING TOTALLY RAMIFIED DEGREE Q EXTENSIONS WITH CONTROLLED UNIFORMIZER RELATIONS IN POSITIVE RESIDUAL CHARACTERISTIC

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Allowed standard prerequisites (editorial).

- Adjoining a q-th root of the uniformizer gives a totally ramified degree q separable extension with the required uniformizer relations. invoked at proof lines 34654-34657. Handles the equal-characteristic-zero case.

Source reference. Original lemma 09EV: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33954–33954. Cited at proof lines 34655-34656 for the characteristic-zero case; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a discrete valuation ring with uniformizer π. If the residue characteristic of A is $p > 0$, then for every $n > 1$ and p-power q there exists a degree q separable extension L/K totally ramified with respect to A such that the integral closure B of A in L has ramification index q and a uniformizer π_B such that $\pi_B^q = \pi + \pi^n b$ and $\pi_B^q = \pi + (\pi_B)^{nq} b'$ for some $b, b' \in B$.*

Proof. If the characteristic of K is zero, then we can take the extension given by $\pi_B^q = \pi$, see Lemma 09EV. If the characteristic of K is $p > 0$, then we can take the extension of K given by $z^q - \pi^n z = \pi^{1-q}$. Namely, then we see that $y^q - \pi^{n+q-1} y = \pi$ where $y = \pi z$. Taking $\pi_B = y$ we obtain the desired result. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.